Escaping the Data-Limited Frontier: Convergence-Bias, Fused-Kernel, and Memorizer-Regularization Levers for the 5-Minute H200 Frame

Fable 5 (hypothesis author), OPHIS [1]

Abstract

Block hypothesis paper for rounds 70+ of the walltime_5min_h200 challenge (minimize val_bpb after 300 s on one H200; reference baseline 1.0301, current SOTA 0.9506 mean / 0.9483 best, single-seed $\sigma \approx 0.0015$, 2σ gate ≈ 0.003). The prior 69-round campaign did not merely exhaust cheap levers; it diagnosed why the frontier is walled: the floor is data-limited. A data-scaling ablation shows a -0.118 val_bpb swing from 5→10 training shards — larger than the entire $+0.082$ from 64 rounds of architecture search — at constant step count; the model trains on 10 of 6543 shards (∼2 epochs) on the overfitting boundary. The n-gram hash memory is a memorizer and the sole carrier of that data-sensitivity (off: range 0.004; on: 0.118; helps -0.029 at 10 shards, harms $+0.086$ at 5), and every structural subtraction lever (FFN, span) inverts with data. We run a protocol-compliant multi-agent idea round (three proposers + an adversarial critic) and select levers on three axes: (a) fixed-inductive-bias convergence levers with ∼zero memorizing capacity (should not invert with data); (b) the never-written fused n-gram-VE kernel and an FA3→FA4 / fp8-attention upgrade to raise the step budget; (c) regularizing the memorizer to generalize at ∼2 epochs. Every lever carries a data-invariance falsifier (run at 8 and 10 shards) and, for architectural levers, a kernel-maturity rule (judge per-update quality first; a slow first kernel is a refinement task, not a rejection).

1. Introduction

The 5-minute frame rewards quality-per-step and steps-per-300s at once, so most levers fail — and here they did, but the informative result is structural: profiling plus a late data-scaling ablation showed the frontier is walled by data, not architecture. The disciplined response is to stop turning capacity knobs (whose verdicts invert with data) and attack the three things that can move a data-limited, memorization-driven, underfit-per-step model. Candidates were generated with the protocol's multi-agent procedure: Proposer A (convergence/architecture), B (throughput/kernel), C (memorizer-regularization), and a Critic that adversarially ranks the pooled slate; the coordinator verifies every claim against the SOTA code.

2. Related Work

Canon layers (Allen-Zhu, 2025): zero-init causal depthwise convolutions summing neighbor hidden states improve loss at negligible params (motivates canon_conv, orthogonal to the discrete memorizer). Differential Transformer (Ye, 2024): dual-softmax subtraction cancels common-mode attention noise. MSWA (MSWA, 2025): diverse window sizes beat uniform SWA. FlashAttention (Dao, 2022) and FA4 (2025, confirmed in Recursive's reported results): attention-kernel/precision upgrades; Recursive's big lever is fp8 attention projections. Data Movement Is All You Need (Ivanov, 2021): fusing index/elementwise ops removes launch overhead (motivates the fused n-gram kernel). Muon NS acceleration (Muon-NS, 2026): comparable orthogonalization in ∼10 vs 15 matmuls. Scaling Data-Constrained LMs (Muennighoff, 2023): ∼4 epochs of repeat ≈ fresh (scope tension: we overfit at 3). Tensorizing Engram (Engram, 2026): shared latents across n-gram tables (motivates factorized codes).

[1] Autoresearch reimplementation campaign. Correspondence to: Fable 5 (OPHIS) <baiyuzhu@mit.edu>.

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3. Candidate levers and five-dimension scoring

Scores 1–10 on novelty/validity/impact/reliability/provenance. Table 1 summarizes the slate. Every quality lever carries a data-invariance falsifier: a 10-shard win that shrinks/inverts at 8 shards is an epoch-trade and is rejected.

Convergence/architecture (A). canon_conv adds $x \mathrel{+}= \mathrm{DWConv}(\mathrm{norm}(x))$ per block (zero-init $\Rightarrow$ no-op at step 0), a differentiable local-composition prior orthogonal to the lookup memorizer, so its verdict should not invert with data. diff_attention and mswa tune attention, which is $\sim$5% of step time.

Throughput/kernel (B) + FA4/fp8 (operator-elevated). The measured $\sim$194k tiny fill kernels sit on the n-gram-VE path; indices are a pure function of tokens+primes, so the path is fusible byte-identically (B1 compile-fuse; B2 Triton/offline). Attention is only $\sim$5% of step time now, so an attention-kernel win (FA4, or fp8-attention which we previously retracted, possibly prematurely) is masked until B1/B2 fuse the n-gram sink; it is therefore sequenced after a fresh re-profile that unmasks it.

Memorizer-regularization (C) — the on-thesis track. All three assume the harm is concentrated in low-count/high-magnitude rare keys; the shared deliverable is the $\{5, 8, 10\}$-shard slope. C1 decoupled WD on the n-gram table alone (continuous count-adaptive backoff). C2 product-code/factorized n-gram (Tensorizing Engram): $\sim$256$\times$ fewer params, coverage preserved — the one lever that could raise the scored 10-shard operating point. C3 frequency-confidence gate $ng \cdot f(c)$ (neural Good-Turing), intrinsically data-adaptive and the cheapest premise-test.

4. Experimental setup and three rigor protocols

Frame walltime_5min_h200, 300 s, one H200, fa3→fa4 under test, max-autotune. GPU-controlled paired same-seed A/B (an fp8 "win" was once a placement artifact). (1) Data-invariance (quality levers): re-run adopted levers at 8 shards; a shrinking/inverting win is an epoch-trade $\Rightarrow$ reject. (2) Two-run throughput (kernel levers): fixed-step byte-identity, then fixed-300s conversion; if steps rise but val_bpb is flat at the 2-epoch boundary, redirect the reclaimed compute into quality, not epochs. (3) Kernel-maturity (architectural levers): judge per-update quality first — a new component's naive kernel may regress step-time without the mechanism being bad; if fixed-step quality clears the gate, a slow kernel is a refinement task, not a rejection (only a null per-update quality kills the mechanism).

5. Results to date and first cycle

Prior campaign: architecture floor certified (+7.9%) then exhausted; floor is data-limited; n-gram memory is the memorizer; kernel bottleneck measured but never fixed. Cycle 1 (B1, this block): the pad-based byte-identical n-gram hash is throughput-null (1938 vs 1948 steps, neutral), so the $\sim$194k fill-kernels are not the hash torch.zeros (Inductor already fuses them); B2 is downgraded pending a fresh re-profile to locate the real sink. B1's cheap null is exactly its purpose: settle the throughput question before investing in B2.

6. Execution plan (critic-finalized)

Rotation avoids same-direction back-to-back; explore:refine $\approx$3:2. Order: (1) B1 fused-compile [done, null]; (2) C3 backoff-gate (premise test, $\{5, 8, 10\}$ slope); (3) A1 canon_conv (fixed-step quality first + 8-shard invariance); (4) C2 factorized codes — highest value (neutral-or-better @10sh AND clears 2σ @5sh); (4b) re-profile + FA4/fp8-attention (after B1/B2 unmask it); (5) B2 (only if a re-profiled sink converts); (6) C1 (sparse-grad-gated). Each cycle runs the full registry chain: mechanism→hypothesis→gated experiment→run→evidence→belief→decision→chart, snapshot on SOTA. A deep-analysis paper is written every five rounds.

References

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Table 1. Multi-agent candidate slate. C = critic verdict (KEEP/CUT + final rank).

Lever	Mechanism (1-line)	n/v/i/r/p	Critic
A1 canon_conv	zero-init causal depthwise conv; orthogonal to memorizer, no-op at init	6/8/6/8/9	KEEP #3
A2 diff_attention	dual-softmax common-mode cancel; attn only 5% of step	7/7/6/5/8	CUT (attn not bottleneck
A3 mswa	graded multi-scale windows	5/6/4/5/7	CUT (span-inversion adja
B1 fused-compile	@torch.compile-fuse hash+gather, byte-identical	5/9/7/8/9	KEEP #1
B2 fused-triton	hand Triton hash→gather→sum / offline idx	7/9/8/6/9	KEEP #5 (cond. on B1)
B3 muon_ns_reduce	fewer NS matmuls; not byte-identical	6/5/4/6/6	CUT (optimizer not bott
FA4/fp8-attn	FA4 upgrade + fp8 attention projections	6/7/7/6/8	operator-elevated #4b
C1 table-only WD	decoupled WD on n-gram table alone	5/8/6/8/7	KEEP #6 (sparse-grad g
C2 factorized codes	product-code n-gram (Tensorizing Engram)	8/7/8/5/8	KEEP #4 (highest value
C3 backoff gate	$ng \cdot f$(count) neural Good-Turing	7/8/7/5/8	KEEP #2 (premise test)